CHIRASMITA GHOSH

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EXPERIENCE

- **Telecom Technology Intern - BSNL** (June 2026 – July 2026)
  - Analyzed and diagnosed 2G/3G/4G telecom infrastructure, executing network routing workflows and signal-path troubleshooting in collaboration with cross-functional engineering teams to ensure system uptime.
- **Business Development Executive Intern - Reddensoft Info. Pvt. Ltd.** (May 2026 – June 2026)
  - Assisted in business development initiatives and client engagement. Worked with cross-functional teams to improve customer outreach and communication. Worked on bidding and confirmation of project deals.

EDUCATION

- **Bachelor of Technology ECE (Ai and Cybernetics)** May 2023 – May 2027
  Vellore Institute of Technology **Grade: 8.72**

SKILLS AND INTERESTS

- **Programming Languages:** C, C++, Embedded C, Python, MATLAB, Verilog, VHDL
- **Digital Design:** Computer Architecture, Digital Logic Design, Digital Circuits, RTL, FPGA Interfacing
- **Embedded Systems:** Microcontrollers, Microprocessors, Arduino, Raspberry Pi, ESP32, RTOS
- **Communication Protocols:** UART, SPI, I2C, CAN, Ethernet, Serial Communication
- **Verification & Testing:** Design Verification, Functional Testing, Testbench, Performance Analysis
- **VLSI & EDA Tools:** Cadence Virtuoso, LTspice, EDA Tools, MATLAB/Simulink, Synthesis Fundamentals
- **Software Development:** Data Structures & Algorithms, Algorithm Development, Software Design, SDLC

PROJECTS

- **Fire Fighting Drone**
  - Developed a real-time computer vision model using Python and OpenCV to detect localized fire hazards and thermal anomalies with high positional accuracy.
  - Integrated the vision system with an onboard edge microprocessor (Raspberry Pi) to process live video feeds directly on the drone, eliminating network latency.
  - Engineered automated triggering logic that interfaces the AI model with hardware relays, automatically actuating a servo-driven fire-suppressant valve upon target lock.
- **Algonaught-Board**
  - Engineered an end-to-end industrial IoT architecture deploying Raspberry Pi sensor nodes to harvest real-time vibration and thermal metrics from heavy machinery.
  - Developed a Raspberry Pi edge gateway that aggregates sensor data via MongoDB, serving as the localized hardware host for the Algonaught-Board deep learning model.
  - Implemented on-device anomaly detection using PyTorch on the Raspberry Pi to predict mechanical failure risks locally, reducing cloud bandwidth usage by 60%.
- **Ultrasonic Radar System**
  - Designed an ultrasonic radar system using an ESP32-WROOM-DevKitV1, HC-SR04 ultrasonic sensor, and servo motor to detect and track nearby objects in real time.
  - Configured GPIO interfaces for trigger, echo, and servo control, integrating embedded C/C++ logic with sensor data acquisition and actuator positioning.
  - Developed a MATLAB-based visualization interface to display object distance and angular position, enabling real-time monitoring, system testing, and hardware validation.

CERTIFICATIONS

- VLSI Chip Design and Simulation for Industrial Applications – LT Edutech
- Machine Learning Specialization – DeepLearning.AI, Stanford University
- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional

EXTRA CURRICULARS

- **Program Representative** – ECE (Ai and Cybernetics) Students' Council, Academic Year 2024–2025
- **Technical Team Lead, IOT Club** – Volunteered for various workshops, led various sessions, and helped create a competitive environment by encouraging students to participate in technical events. (2024–2025)
- **Floor Student Representative** – Hostel Administrative Council